

Yan Kevin ZE

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PROFILE

Student at the African Institute for Mathematical Sciences (AIMS) Rwanda. Passionate about learning, developing, and teaching mathematics, I am seeking opportunities to strengthen my mathematical skills, conduct research, and contribute to the community. My research interests lie in Algebra, Topology, and Geometry.

EDUCATION

African Institute for Mathematical Sciences (AIMS)

Kigali, Rwanda

MSc in Mathematical Sciences | 2025–Present

Thesis: On Stallings' Fibration Theorem for Knot Exteriors.

Supervisor: Professor Ana Garcia Lecuona

Study of Stallings' fibration theorem for knot exteriors, connecting the algebraic structure of the knot fundamental group to fibrations over S^1 via Seifert surfaces and infinite cyclic coverings.

University of Yaoundé I

Cameroon

Master 2 in Mathematics (Geometry) | 2023–2024 | Status: Awaiting defense

Thesis: (Co)homology of Some Homogeneous Spaces: The Case of Stiefel and Complex Grassmann Manifolds.

Supervisor: Dr. Hilaire Mbiakop Georges

Computation of singular cohomology of Stiefel manifolds $V_{k,n}$ and complex Grassmannians $G_{k,n}$, using algebraic topology tools including the Gysin exact sequence, Euler and Chern classes.

University of Yaoundé I

Cameroon

Master 1 in Pure Mathematics (Algebra and Geometry) | 2022–2023

University of Yaoundé I

Cameroon

Bachelor's Degree in Fundamental Mathematics | 2021–2022

Baccalauréat C

Cameroon

2018–2019

RESEARCH EXPERIENCE

MIORPA Research Programme — University of Oxford

Oxford, United Kingdom

Research Intern | July – September 2026

Project: New supergravity solutions

Supervisors: Tabea Sieper & Chris Couzens

Study of new supergravity solutions of the form $AdS_3 \times M(7)$ encoding the near-horizon region of a black hole, using toric geometry methods and differential equations in Mathematica.

TECHNICAL SKILLS

Programming: R, LaTeX, Python

Software: Microsoft Word, Excel, Mathematica

LANGUAGES

French: Fluent **English:** Working proficiency **Bikélé:** Mother tongue, fluent

RELEVANT COURSES

Topology and Series; Functional Analysis; Dynamical Systems I and II; Topological Vector Spaces; Lie Groups and Lie Algebras; Groups, Rings, Modules, and Algebras I; Homological Methods; Linear and Bilinear Algebra; Riemannian Geometry; Symplectic Geometry I and II; Differential Geometry I and II; Algebraic Geometry I; Algebraic Topology; De Rham Cohomology.

CONFERENCES AND WORKSHOPS

Mathematics Conference

Theme: Recent Developments in Riemannian and Contact Geometry.

Workshop on Algebra, Logic and Geometry (AALG-24)

Theme: Algebro-Geometric Methods for Information Systems.

Mathematical School: Algebra and Geometry

Theme: Scientific Production with LaTeX.

TEACHING EXPERIENCE

University of Yaoundé I

Cameroon

Tutor | 2023, 2024

- Taught Linear Algebra I and II.
- Taught Ordinary Differential Equations.

High School

Cameroon

Tutor | 2018, 2022, 2023, 2024

- Taught Mathematics and Physics.
- Levels: Terminale C and Première C.

AWARDS

- 2023–2024: Presidential Excellence Award
- 2015–2016, 2017–2018: Best High School Student Award

LEADERSHIP AND VOLUNTEERING

Volunteer — OneChild OneDream

- Participated in fundraising activities to support visits to orphanages.

HOBBIES

Reading, Meditation, Martial Arts (Kung-Fu), Tennis